Device-wise Conformal False-Alarm Calibration for Federated IoT Intrusion Detection

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Abstract. Federated learning (FL) enables data-local intrusion detection across distributed IoT devices, but average detection metrics can hide severe device-level false-alarm disparity: a single global alarm threshold may achieve the desired fleet-level false-positive rate (FPR) while producing almost no false alarms on some devices and excessive false alarms on others. We formulate this reliability gap as a *device-wise false-alarm risk control* problem and study **PFC-FedIDS**, which trains a shared FL backbone, performs benign-only split conformal calibration per device, and optionally keeps parameter-efficient bottleneck adapters local. Our contribution is an operational formulation and an empirical study rather than a new conformal algorithm: the reliability mechanism is the per-device threshold, which targets a user-specified FPR without uploading device-specific parameters or requiring labeled attacks for calibration, and we show that the adapter is neither necessary nor safe. Across three seeds on N-BaIoT (9 devices) and Edge-IIoTset (10 heterogeneous sensors), global thresholding matches the average FPR but produces worst-device gaps of 22.8% on Edge-IIoTset, versus 1.7–2.1% for device-wise calibration at equal or better mean detection (TPR 0.9575 vs. 0.9528). We then stress-test the exchangeability assumption behind the guarantee. Device-wise calibration keeps its advantage under chronological splits, exact-duplicate removal, restored attack classes and held-out attack types, with a consistent ordering in every seed, but the margin is protocol-dependent: combining a chronological split with duplicate removal shrinks the gap reduction from roughly 11× to 1.6× and pushes one sensor to 13.4% FPR against a 3% target. The conformal statement is a per-device marginal upper bound, not simultaneous fleet control, and it does not survive non-exchangeable deployment conditions without recalibration.

Keywords: Federated Learning · Intrusion Detection · Conformal Prediction · IoT Security · False-Alarm Control

1 Introduction

The Internet of Things (IoT) ecosystem spans diverse devices—cameras, thermostats, doorbells, baby monitors, industrial sensors—each generating distinct network traffic patterns. Deploying a single intrusion detection system (IDS)

across this heterogeneous fleet faces a fundamental reliability challenge: benign traffic distributions differ substantially across device types, so a detection threshold appropriate for one device may trigger excessive false alarms on another [1].

Federated learning (FL) supports data-local IoT intrusion detection by training models locally on each device and aggregating updates on a central server [2,3]. Recent work has applied FL to IoT IDS [4,5], explored personalized FL for heterogeneous clients [6,7], and studied federated calibration and conformal prediction [11,12]. However, existing work either (1) optimizes average detection accuracy without addressing per-device false-alarm disparity, or (2) studies calibration in general FL settings without grounding it in the concrete reliability requirements of IoT intrusion detection.

Problem gap. Consider Edge-IIoTset, with 10 sensor types. Under standard FedAvg with a global conformal threshold targeting $\alpha = 3\%$ FPR, the fleet-level average FPR is close to target, yet the mean worst-device FPR gap across three seeds is 22.8%. The average metric therefore gives a false sense of reliability: some devices are under-sensitive while others generate many false alarms. For a security operator this is unacceptable because alarm load and trust are experienced per device, not in aggregate. The core problem is *device-level false-alarm risk heterogeneity*.

Our approach. We study **PFC-FedIDS** (Personalized Federated Conformal Intrusion Detection System), which reformulates federated IoT IDS as a device-wise false-alarm risk control problem. The pipeline has three components: (1) a shared encoder-classifier backbone trained via FedAvg, (2) benign-only device-wise split conformal calibration that assigns each device a personalized alarm threshold targeting a user-specified FPR α, and (3) an optional parameter-efficient local bottleneck adapter. Component (2) is the reliability mechanism: threshold calibration must respect device-specific benign score distributions, because pooling benign calibration data can preserve the average FPR while destroying per-device control. Component (3) is reported as an ablation axis rather than a contribution; our measurements show it provides no reliable benefit on these benchmarks and introduces a rank-dependent failure mode.

Contributions. We make three contributions:

1. **Reliability formulation**: We formulate federated IoT intrusion detection as device-wise false-alarm risk control and argue that average FPR is insufficient for heterogeneous IoT deployments.
2. **Operational study, not a new algorithm**: We evaluate benign-only device-wise conformal FPR control for federated IDS, specify the finite-sample threshold exactly, and separate threshold personalization from model personalization. Both ingredients are known; the contribution is the operational formulation, the protocol, and the evidence.
3. **Empirical evidence and its limits**: Across two device-structured datasets and three seeds, global thresholding matches the average FPR while producing large device-level gaps, whereas device-wise calibration tracks target FPRs across target values and calibration budgets. We additionally stress-test the exchangeability assumption and report the protocol under which

the per-device budget is empirically violated, together with a paired analysis showing that the adapter's effect is not statistically distinguishable.

2 Related Work

FL for IoT intrusion detection. FL-based IDS is a mature area: Campos et al. [4] evaluated FL for IoT IDS on ToN-IoT and Vyas et al. [5] surveyed the field. We claim no novelty in applying FL to IDS and instead address one unresolved problem: device-level false-alarm disparity.

Personalized FL. FedAvg [2] and FedProx [3] are the foundational algorithms, the latter addressing statistical heterogeneity via a proximal term. Personalized FL adds client-specific components: prototype-based personalization [6], knowledge distillation for heterogeneous IoT [7], and graph-driven personalization on N-BaIoT [8]. Our optional adapter follows parameter-efficient fine-tuning (LoRA [9]) but keeps adapters local.

Conformal prediction and federated calibration. Conformal prediction gives distribution-free uncertainty quantification [10]. FedCal [11] studies local and global calibration via temperature scaling; Federated Conformal Predictors [12] extend conformal prediction to FL under exchangeability violations; Personalized Federated Conformal Prediction [13] targets agent-specific uncertainty sets; CF-HFC [14] combines calibrated FL with adaptive conformal calibration for heterogeneous IoT IDS.

Why we do not report these as numerical baselines. FCP and PFCP output prediction sets or agent-specific uncertainty regions and target set coverage, not an operator-specified false-alarm budget estimated from benign traffic only; mapping them onto benign-only score thresholding would require a surrogate objective that neither paper defines, so the resulting numbers would measure our surrogate rather than their method. CF-HFC is closest in setting but centers on hardware-aware fuzzy clustering and has no available implementation. We therefore compare structurally and isolate a failure mode none of them study: pooled calibration can match average FPR while violating per-device operation, and model personalization alone does not remove it.

Our positioning. Our contribution is not a new conformal algorithm. It is to (1) formalize *device-wise false-alarm disparity* as an operational reliability metric for federated IoT IDS, (2) show empirically that global calibration can match the average FPR while failing device-level control (22.8–23.7% FPR gap), and (3) evaluate a benign-only per-device calibration pipeline across two device-structured benchmarks, including stress tests that identify when the underlying exchangeability assumption breaks.

3 Problem Formulation

Consider K IoT devices, each acting as a federated client. Device i holds a local dataset $D_i = \{(x_j^i, y_j^i)\}_{j=1}^{n_i}$ where $x_j^i \in \mathbb{R}^d$ are traffic features and $y_j^i \in \{0, 1\}$ (0 = benign, 1 = attack). Benign traffic distributions $P_i(x \mid y = 0)$ differ across

devices due to heterogeneous device types, communication protocols, and usage patterns.

Standard FL objective. The conventional FL goal is to minimize the average empirical risk across clients:

$$\min_{\theta} \sum_{i=1}^{K} \frac{n_i}{N} \mathcal{L}_i(\theta) \tag{1}$$

where $\mathcal{L}_i(\theta) = \frac{1}{n_i} \sum_j \ell(f_\theta(x_j^i), y_j^i)$ and $N = \sum_i n_i$.

Our objective. We argue that for trustworthy IoT IDS, minimizing average loss is insufficient. Instead, we aim to:

$$\min_{\theta, \{\phi_i\}, \{q_i\}} \sum_{i=1}^{K} \frac{n_i}{N} \mathcal{L}_i(\theta, \phi_i) \quad \text{s.t.} \quad \mathrm{FPR}_i(q_i) \le \alpha, \ \forall i \tag{2}$$

where ϕ_i are device-specific adapter parameters kept locally, q_i is a device-specific alarm threshold, and α is the target FPR set by the security operator. We use the constraint as a design objective for per-device false-alarm operation. The conformal guarantee supporting it is a marginal per-device upper bound under the assumptions in §4.2, not simultaneous control over all devices and not conditional control for each realized device stream.

Key metrics. Beyond standard detection metrics (F1, TPR), we measure cross-device FPR dispersion FPR-Std = $\mathrm{Std}(\mathrm{FPR}_1, \ldots, \mathrm{FPR}_K)$, worst-case disparity FPR-Gap = $\max_i \mathrm{FPR}_i - \min_i \mathrm{FPR}_i$, the corresponding TPR-Std and worst-device sensitivity $\min_i \mathrm{TPR}_i$, and the per-device deviation from target $|\mathrm{FPR}_i - \alpha|$.

4 Method: PFC-FedIDS

The core pipeline combines federated backbone training, server aggregation, and DeviceCP; the local bottleneck adapter is an optional personalization layer.

4.1 Shared Encoder with Local Bottleneck Adapters

The model has the structure:

$$z = \mathrm{Encoder}_\theta(x); \quad z_i' = z + \mathrm{Adapter}_{\phi_i}(z); \quad \hat{y} = \mathrm{Classifier}_\theta(z_i') \tag{3}$$

The encoder and classifier parameters θ are shared across all devices and aggregated on the server. Each adapter is a bottleneck module:

$$\mathrm{Adapter}_{\phi_i}(z) = W_{up}^i \cdot \mathrm{ReLU}(W_{down}^i \cdot z) \tag{4}$$

where $W_{down}^i \in \mathbb{R}^{r \times d_h}$, $W_{up}^i \in \mathbb{R}^{d_h \times r}$, and $r \ll d_h$ is the adapter rank (we use $r = 16$, $d_h = 128$). The adapter has only $2 \cdot r \cdot d_h$ parameters (4,096 for

our configuration, which is 6.0% of the N-BaIoT model and 8.8% of the smaller Edge-IIoTset model) and is never uploaded to the server.

Training protocol. Each round, the server distributes the global θ; each client trains the full model (including its local adapter) for E epochs, optionally with a FedProx proximal term, and uploads only θ; the server aggregates by FedAvg weighted by local dataset sizes. Adapters remain local throughout. This data-local and parameter-local design should not be read as a formal privacy guarantee.

4.2 Device-wise Conformal Calibration

After FL training, each device holds a calibration set $D_i^{cal} = \{(x_j, y_j = 0)\}$ of *benign-only* samples (never used in training). We define the anomaly score as:

$$s(x) = 1 - p_{\theta,\phi_i}(\text{benign} \mid x) \tag{5}$$

For device i, let $s_{(1)} \leq \cdots \leq s_{(n_i^{cal})}$ be the sorted benign calibration scores and augment them with $s_{(n_i^{cal}+1)} := +\infty$. The conformal rule is [10]:

$$r_i = \left\lceil (n_i^{cal} + 1)(1 - \alpha) \right\rceil, \qquad q_i = s_{(r_i)}. \tag{6}$$

The second branch is active when $n_i^{cal} < (1 - \alpha)/\alpha$. In that regime no finite threshold obtained from the calibration scores can certify the requested level, so the safe fallback raises no alarms. All reported experiments have sufficient benign calibration data for the finite first branch. We use the exact order statistic rather than an interpolated empirical quantile, because interpolation between adjacent order statistics does not correspond to the finite-sample argument below.

At test time, the alarm decision for a sample x on device i uses the strict inequality:

$$\text{Alarm}_i(x) = \begin{cases} 1, & s(x) > q_i \\ 0, & s(x) \leq q_i \end{cases} \tag{7}$$

The inequality is strict so that calibration scores tied with q_i do not raise an alarm; ties therefore make the realized FPR smaller than α rather than larger, which preserves the upper bound at the cost of detection power.

Why benign-only? In IoT security operations benign traffic is abundant while attacks are scarce and unknown a priori, so an operator can calibrate from a few hours of normal traffic collected during a trusted window without any labeled attacks.

Theoretical guarantee. Under the within-device exchangeability assumption (samples in D_i^{cal} and benign test samples are exchangeable), split conformal prediction provides the marginal upper-bound guarantee:

$$\Pr(s(X) > q_i \mid Y = 0) \leq \alpha \tag{8}$$

where q_i is the order statistic of Eq. (6). The $+1$ in $(n_i^{cal} + 1)(1 - \alpha)$ is the finite-sample correction that makes the rank of q_i valid for the unseen test point

rather than only for the calibration sample. With n_i^{cal} calibration samples, the expected FPR is conservatively bounded:

$$\mathbb{E}[\mathrm{FPR}_i] \leq \alpha \tag{9}$$

This is a *marginal* guarantee: it controls the per-device FPR averaged over the randomness of both calibration and test samples. It is not conditional on a realized calibration set, covariate subgroup, or device state, and it is not a simultaneous high-probability guarantee over all devices in a deployment. It also does not guarantee closeness to α from below; with discrete scores or heavy ties, the empirical FPR can be substantially smaller than α (conservative). In practice, with several hundred benign calibration samples and reasonably well-separated score distributions, the empirical FPR approaches the target, as shown in the calibration-size sensitivity study. Global calibration over pooled benign traffic does not provide this per-device guarantee when devices have heterogeneous score distributions.

5 Experiments

5.1 Datasets and Setup

We evaluate on two datasets with device-structured federated partitions:

N-BaIoT [1] contains network traffic from 9 commercial IoT devices (doorbells, cameras, thermostats, baby monitors) infected with Mirai and BASHLITE botnets. Each device naturally forms one federated client (9 clients). We use 115-dimensional traffic statistics as features and binary classification (benign vs. attack).

Edge-IIoTset [15] contains network traffic from 10 heterogeneous IoT/IIoT sensors (distance, flame, heart rate, IR receiver, Modbus, pH, soil moisture, sound, temperature/humidity, water level) under 14 released attack types across 5 categories (DDoS, MITM, injection, malware, reconnaissance). Each sensor type defines one client for benign traffic, while attack traces are randomly assigned across clients to simulate per-device attack exposure. This preserves real sensor-specific benign heterogeneity but uses synthetic attack assignment. We use 33 common numeric features from protocol headers (TCP, UDP, DNS, MQTT, Modbus, ICMP, HTTP, ARP).

Attack coverage of our Edge-IIoTset partition. Our pipeline coerces the 33 numeric columns and drops rows with non-finite values. Because the released CSVs store `icmp.checksum` as hexadecimal strings in some files, this silently removes all `DDoS_ICMP_Flood`, 58% of `OS_Fingerprinting` and 31% of `MITM` records, so our benchmark covers 13 of the 14 released attack classes and the missing one is the most frequent DDoS variant. Section 5.9 quantifies the effect: restoring these records leaves the device-wise calibration conclusion unchanged but lowers detection under a pooled global threshold.

Both datasets are split per device into train (60%), calibration (20%), and test (20%) sets with stratification by label, and standardization is fitted only

on training partitions. This record-level random split is the primary benchmark protocol; §5.9 replaces it with chronological and duplicate-free variants.

Per-device calibration budget. The benign calibration records available per device set the granularity of the attainable FPR. On N-BaIoT this ranges from 2,623 to 4,000 (median 4,000), so even the smallest device supports an FPR resolution of 0.04% ($k = 2{,}546$ at $\alpha = 3\%$); on Edge-IIoTset every device has exactly 3,000 records ($k = 2{,}911$), so the 3,000-sample point in the calibration-budget study is that benchmark's full budget rather than an arbitrary cut-off.

5.2 Baselines and Variants

We compare 11 methods across three axes—FL backbone, thresholding strategy, and personalization mechanism. **Local MLP** trains each device independently; **Centralized MLP** pools all data (pooled-data reference). **FedAvg+DeviceCP** [2] and **FedProx+DeviceCP** [3] ($\mu = 0.01$) combine a shared MLP backbone with per-device conformal thresholds, while **FedAvg+GlobalCP** replaces those with one pooled threshold. **FedPer+DeviceCP** and **FedPer+GlobalCP** [17] use a shared encoder with local per-device classifiers, which separates model personalization from threshold personalization. **PFC-FedIDS** (ours) adds local adapters ($r = 16$) to the shared backbone with per-device calibration ($\alpha = 0.03$); its ablations remove the adapter (**FedAvg+DeviceCP w/o Adapter**, the strong calibration baseline), remove calibration (**w/o Conformal**, argmax at threshold 0.5), or replace per-device with pooled thresholds (**GlobalCal**).

5.3 Implementation Details

All methods use a 2-hidden-layer MLP backbone ($115/33 \to 256 \to 128 \to 2$) with BatchNorm, ReLU, and Dropout(0.2). For PFC-FedIDS, a bottleneck adapter (rank $r = 16$) is inserted after the 128-dim hidden layer. Training uses Adam optimizer ($lr = 10^{-3}$), batch size 256, 50 communication rounds, and 1 local epoch per round. We use 40,000 samples per device for N-BaIoT and 30,000 samples per device for Edge-IIoTset, balanced between benign and attack traffic. Conformal target FPR is $\alpha = 0.03$ unless otherwise stated. We report mean $\pm$ population SD (NumPy `ddof=0`) over training seeds 42/43/44 on one fixed partition per protocol; these runs quantify training stochasticity only. All experiments run on a 6-core CPU workstation without GPU acceleration; training is deterministic under a fixed seed, and every number in Tables 1 and 2 was re-generated from scratch and matched exactly during a reproducibility audit.

5.4 Evaluation Protocol and Research Questions

We report a **dual-protocol** evaluation: (1) standard argmax (threshold=0.5) for detection capability (F1, TPR), for comparison with prior FL IDS work; (2) conformal thresholds from benign calibration sets for false-alarm control

(FPR-Std, MaxFPR-Gap, $|\text{FPR} - \alpha|$). We also report TPR and F1 under the conformal protocol to verify that device-wise calibration does not degrade attack sensitivity. The "w/o Conformal" variant reports identical values across both protocols; GlobalCP and GlobalCal use a single threshold computed from pooled calibration benign data.

Our evaluation asks five questions. **RQ1**: can global thresholding control device-level false alarms, or only the fleet average? **RQ2**: does device-wise calibration track different target FPR values? **RQ3**: how much benign calibration data is needed? **RQ4**: what is the role of the local adapter relative to calibration? **RQ5**: does model personalization alone eliminate device-level FPR disparity, or is threshold personalization separately necessary?

5.5 Main Results: Device-Level False-Alarm Control

Table 1 and Table 2 present results on N-BaIoT and Edge-IIoTset respectively.

Key observations for RQ1, RQ4, and RQ5:

1. **Average FPR is misleading.** On Edge-IIoTset, FedAvg+GlobalCP achieves an average conformal FPR of 0.0267 ± 0.0016, close to the target $\alpha = 0.03$, yet its MaxFPR-Gap is 0.2280 ± 0.0075. Global thresholding controls the fleet average but fails device-level control.
2. **Attack detection is preserved on average.** On Edge-IIoTset, PFC-FedIDS reaches TPR_{conf}=0.9575 ± 0.0012 and F1_{conf}=0.9637 ± 0.0007 versus 0.9528 ± 0.0010 and 0.9635 ± 0.0012 for FedAvg+GlobalCP; on N-BaIoT TPR_{conf} is effectively 1.0 for all methods. The w/o Conformal variant attains higher argmax-F1 but has no calibrated false-alarm bound, and GlobalCal degrades TPR to 0.9315 because the single over-threshold suppresses alarms for some sensor types.
3. **Model personalization alone does not fix threshold disparity.** FedPer uses local per-device classifiers, a stronger personalization mechanism than our adapter, yet FedPer+GlobalCP still fails: MaxFPR-Gap 0.1258 ± 0.0294 on N-BaIoT and 0.2518 ± 0.0462 on Edge-IIoTset, comparable to or worse than FedAvg+GlobalCP. A personalized classifier improves detection but does not re-align heterogeneous benign score distributions; only after per-device calibration does FedPer reach the lowest gap in our comparison (0.0087 ± 0.0005). Model and threshold personalization are therefore separate reliability levers.
4. **The adapter is not the source of false-alarm control, and it carries a failure mode.** The no-adapter baseline matches or exceeds the full method on most metrics. On N-BaIoT the adapter also produces a catastrophic single-seed failure: on seed 44 its argmax F1 drops from 0.9995 to 0.9464 (the 0.9817 in Table 1 is the three-seed mean, which averages this collapse with two normal seeds). Section 5.10 isolates the cause. Conformal-protocol F1_{conf}=0.9865 nevertheless remains competitive (vs. 0.9867 for w/o Adapter), because device-wise calibration absorbs most of the variance the adapter introduces. The adapter adds 4,096 local parameters (6.0% of the

Table 1. N-BaIoT results (9 devices) over seeds 42/43/44 on one fixed split. Each cell is the mean with the population SD as a subscript (leading zero omitted). F1/TPR columns report argmax (threshold=0.5) and conformal (Eq. 6, device-wise α=0.03) protocols. The *w/o Adapter* ablation of PFC-FedIDS is by construction the FedAvg+DeviceCP row and is not repeated.

Method	$F1_{arg}\uparrow$	$F1_{conf}\uparrow$	$TPR_{conf}\uparrow$	$FPR\text{-}Std_{conf}\downarrow$	$MaxGap_{conf}\downarrow$
Local MLP	$0.9994_{\pm.0000}$	$0.9868_{\pm.0004}$	$0.9999_{\pm.0000}$	$0.0032_{\pm.0014}$	$0.0113_{\pm.0053}$
Centralized	$\mathbf{0.9997}_{\pm.0000}$	$\mathbf{0.9935}_{\pm.0003}$	$0.9999_{\pm.0000}$	$0.0076_{\pm.0008}$	$0.0243_{\pm.0014}$
FedAvg+DeviceCP	$0.9995_{\pm.0000}$	$0.9867_{\pm.0002}$	$\mathbf{1.0000}_{\pm.0000}$	$0.0041_{\pm.0004}$	$0.0141_{\pm.0004}$
FedProx+DeviceCP	$0.9993_{\pm.0000}$	$0.9861_{\pm.0002}$	$\mathbf{1.0000}_{\pm.0000}$	$0.0036_{\pm.0003}$	$0.0134_{\pm.0013}$
FedAvg+GlobalCP	$0.9995_{\pm.0000}$	$0.9864_{\pm.0002}$	$0.9999_{\pm.0000}$	$\underline{0.0275}_{\pm.0114}$	$\underline{0.0878}_{\pm.0392}$
FedPer+GlobalCP	$0.9993_{\pm.0000}$	$\mathbf{0.9870}_{\pm.0002}$	$\mathbf{1.0000}_{\pm.0000}$	$\underline{0.0412}_{\pm.0073}$	$\underline{0.1258}_{\pm.0294}$
PFC-FedIDS	$0.9817_{\pm.0249}$	$0.9865_{\pm.0001}$	$\mathbf{1.0000}_{\pm.0000}$	$0.0030_{\pm.0007}$	$0.0094_{\pm.0024}$
w/o Conformal	$0.9817_{\pm.0249}$	$0.9817_{\pm.0249}$	$0.9759_{\pm.0339}$	$\mathbf{0.0007}_{\pm.0000}$	$\mathbf{0.0022}_{\pm.0001}$
GlobalCal	$0.9817_{\pm.0249}$	$0.9866_{\pm.0003}$	$\mathbf{1.0000}_{\pm.0000}$	$\underline{0.0438}_{\pm.0051}$	$\underline{0.1379}_{\pm.0258}$
FedPer+DeviceCP	$0.9993_{\pm.0000}$	$0.9866_{\pm.0004}$	$\mathbf{1.0000}_{\pm.0000}$	$0.0045_{\pm.0008}$	$0.0162_{\pm.0027}$

Table 2. Edge-IIoTset results (10 sensors) over seeds 42/43/44 on one fixed split. Cells as in Table 1.

Method	$F1_{arg}\uparrow$	$F1_{conf}\uparrow$	$TPR_{conf}\uparrow$	$FPR\text{-}Std_{conf}\downarrow$	$MaxGap_{conf}\downarrow$
Local MLP	$0.9669_{\pm.0003}$	$0.9670_{\pm.0018}$	$0.9643_{\pm.0037}$	$\mathbf{0.0035}_{\pm.0012}$	$\mathbf{0.0112}_{\pm.0039}$
Centralized	$0.9586_{\pm.0267}$	$0.9667_{\pm.0058}$	$0.9625_{\pm.0113}$	$0.0080_{\pm.0001}$	$0.0307_{\pm.0015}$
FedAvg+DeviceCP	$0.9531_{\pm.0046}$	$0.9632_{\pm.0006}$	$0.9565_{\pm.0009}$	$0.0061_{\pm.0012}$	$0.0207_{\pm.0050}$
FedProx+DeviceCP	$\mathbf{0.9693}_{\pm.0021}$	$0.9624_{\pm.0001}$	$0.9563_{\pm.0007}$	$0.0045_{\pm.0017}$	$0.0142_{\pm.0057}$
FedAvg+GlobalCP	$0.9531_{\pm.0046}$	$0.9635_{\pm.0012}$	$0.9528_{\pm.0010}$	$\underline{0.0677}_{\pm.0022}$	$\underline{0.2280}_{\pm.0075}$
FedPer+GlobalCP	$0.9274_{\pm.0054}$	$0.9267_{\pm.0064}$	$0.8845_{\pm.0084}$	$\underline{0.0753}_{\pm.0139}$	$\underline{0.2518}_{\pm.0462}$
PFC-FedIDS	$0.9510_{\pm.0071}$	$\mathbf{0.9637}_{\pm.0007}$	$\mathbf{0.9575}_{\pm.0012}$	$0.0050_{\pm.0008}$	$0.0174_{\pm.0029}$
w/o Conformal	$0.9510_{\pm.0071}$	$0.9510_{\pm.0071}$	$0.9109_{\pm.0151}$	$0.0076_{\pm.0088}$	$0.0248_{\pm.0301}$
GlobalCal	$0.9510_{\pm.0071}$	$0.9529_{\pm.0022}$	$0.9315_{\pm.0041}$	$\underline{0.0706}_{\pm.0012}$	$\underline{0.2366}_{\pm.0041}$
FedPer+DeviceCP	$0.9274_{\pm.0054}$	$0.9183_{\pm.0323}$	$0.9049_{\pm.0272}$	$\mathbf{0.0025}_{\pm.0001}$	$\mathbf{0.0087}_{\pm.0005}$

N-BaIoT model, 8.8% of the Edge-IIoTset model) and, unlike FedPer's classifier, does not replace the shared head.

5.6 Per-Device FPR Analysis

Figure 1 shows representative per-device conformal FPR for four methods on both datasets. On Edge-IIoTset, FedAvg+GlobalCP exhibits extreme FPR variation: some sensors remain near 0% FPR while another reaches 22.3%. In contrast, PFC-FedIDS keeps all sensors within 1.3–3.4% FPR, much closer to the target $\alpha = 3\%$. On N-BaIoT, global thresholding is also unstable: FedAvg+GlobalCP spans 0.3–14.7% FPR, while PFC-FedIDS spans 2.3–3.5%.

The same representative per-device run also checks whether improved FPR control hides a worst-device detection loss. On Edge-IIoTset, the conformal TPR range is similar for FedAvg+DeviceCP (min TPR 0.842, TPR-Std 0.039), FedAvg+GlobalCP (min TPR 0.880, TPR-Std 0.029), and PFC-FedIDS (min TPR 0.858, TPR-Std 0.035). Thus the main gain of device-wise calibration is not higher worst-device TPR in this benchmark; it is the large reduction in device-level false-alarm disparity while maintaining comparable per-device attack sensitivity. On N-BaIoT, all three methods have min TPR 1.000 in the representative run.

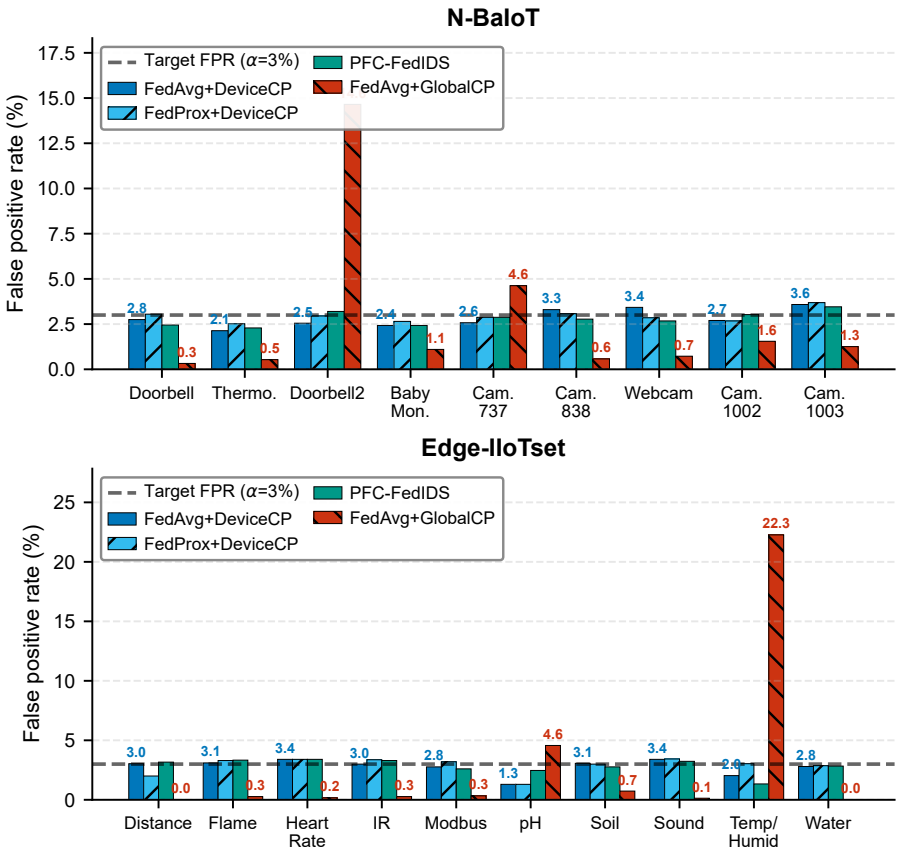


Fig. 1. Seed-42 per-device conformal FPR (one of three runs summarized in Tables 1–2). GlobalCP causes extreme variation; DeviceCP keeps FPRs near $\alpha = 3\%$.

5.7 Ablation and Calibration Comparison (RQ1 and RQ4)

Tables 1 and 2 isolate the source of the gain. Removing conformal calibration eliminates both the user-specified FPR target and the finite-sample statement, whereas FedAvg+DeviceCP captures essentially all of the false-alarm benefit; the adapter is therefore optional, and its instability is analyzed in §5.10. Global calibration is the critical failure mode: on Edge-IIoTset it yields FPR-Std of 6.8–7.1% and MaxGap of 22.8–23.7%, versus 0.50% and 1.7% for device-wise calibration (about a 14× reduction). The full method's MaxGap improvement over GlobalCal is 13.6× under the record-level random protocol but only 1.6× under the hardened protocol (§5.9).

5.8 Robustness Analysis (RQ2 and RQ3)

Figure 2 shows that the device-wise variants remain close across $\alpha \in \{1\%, 3\%, 5\%, 10\%\}$, while GlobalCP MaxGap rises to 45–47% at $\alpha = 10\%$.

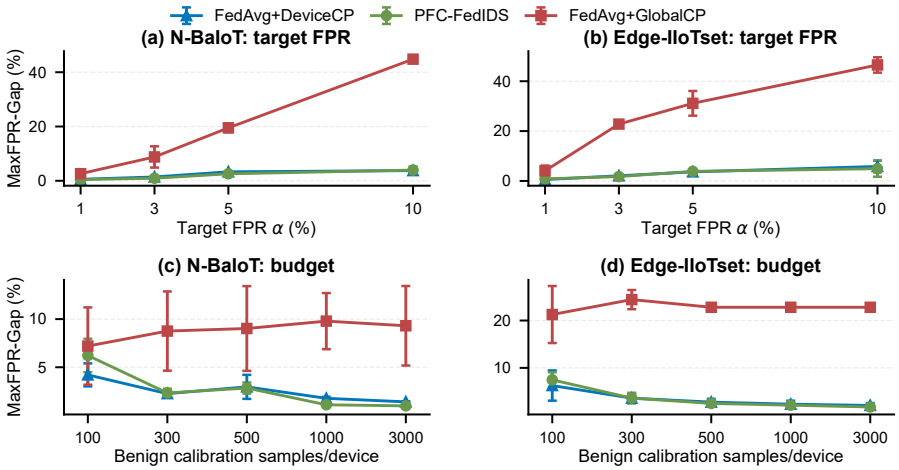


Fig. 2. MaxFPR-Gap sensitivity to target FPR and benign calibration budget (three-seed mean ± population SD).

On Edge-IIoTset, PFC-FedIDS tracks the requested fleet mean (1.01% to 9.78%) and has MaxGap of 1.74% and 4.92% at $\alpha = 3\%$ and 10%, versus 22.80% and 46.52% for GlobalCP. Increasing the benign calibration budget from 100 to 3,000 samples reduces its Edge-IIoTset gap from 7.48% to 1.74%, whereas GlobalCP stays at 21–24%; more calibration data does not repair cross-device score shift.

5.9 Deployment Stress Tests

Eq. (8) assumes within-device exchangeability. Record-level random splitting approximates it for benchmark records but cannot establish it for a future stream. We therefore rebuild Edge-IIoTset from raw captures under alternative protocols while holding features, per-device sample counts, class balance, and train-only standardization fixed.

Chronological split: `frame.time` is parseable for 9 of 10 benign and 12 of 14 attack files, and row order follows capture order (under 0.07% non-monotone steps, with a visible midnight wrap), so we split each device by capture order into the first 60%, next 20% and last 20%. Attack files are split chronologically before assignment, so no attack record crosses a device's time boundary; assignment itself remains synthetic. *Duplicate removal*: the primary random split contains 29% within-partition duplicate rows and 993 unique vectors shared between train and calibration; a chronological split does *not* fix this (1,092 shared vectors, slightly more), because identical vectors recur throughout a capture, so the two mitigations are orthogonal and we also evaluate their combination. *Restored attack classes*: decoding hexadecimal `icmp.checksum` values restores `DDoS_ICMP_Flood` and truncated `OS_Fingerprinting` records, moving the test attack mix from 62% UDP flood to 34% UDP / 27% ICMP. *Held-out attacks*:

Table 3. Edge-IIoTset stress tests over seeds 42/43/44 on one fixed split per protocol, $\alpha = 3\%$. MaxGap is mean ± population SD; TPR and worst-device FPR are three-seed means. DeviceCP (FedAvg+DeviceCP) has smaller MaxGap than GlobalCP (FedAvg+GlobalCP) in every run, but the hardened protocol violates the per-device budget.

	$\text{MaxGap}_{\text{conf}}\downarrow$		$\text{TPR}_{\text{conf}}\uparrow$		DeviceCP worst
Protocol	DeviceCP	GlobalCP	DeviceCP	GlobalCP	device FPR
Record-level random (stress rebuild)	0.0246 ± 0.0022	0.2693 ± 0.0271	0.9673	0.9524	0.039
Chronological split	0.0391 ± 0.0102	0.1893 ± 0.0611	0.9682	0.9428	0.044
Random + duplicates removed	0.0329 ± 0.0006	0.2911 ± 0.0032	0.9685	0.9492	0.050
Chronological + duplicates removed	**0.1051 ± 0.0398**	0.1668 ± 0.0083	0.9731	0.9742	**0.106**
Random + ICMP class restored	0.0199 ± 0.0058	0.2469 ± 0.0656	0.9471	0.8891	0.038
Held-out attack types	0.0123 ± 0.0019	0.2917 ± 0.0065	0.8685	0.7878	0.034

MITM, OS Fingerprinting and Ransomware are excluded from training and calibration.

The ordering is robust; the margin is not. Device-wise calibration yields the smaller worst-device gap in every protocol and every seed (18/18 paired comparisons), but the advantage ranges from 23× (held-out attacks) to 1.6× (chronological + duplicate-free). "An order of magnitude" is thus specific to the record-level random protocol and is not a deployment guarantee.

The per-device budget is empirically violated under the hardened protocol. Under every other protocol no device in any seed exceeds twice the 3% target; under chronological + duplicate-free, 3 of 30 device-seed cells exceed twice the target and 2 exceed three times it, with the heart-rate sensor at 13.4% FPR in two of three seeds while the Modbus client is suppressed to 0.03%. This is the expected consequence of losing exchangeability, measured rather than assumed: the marginal bound does not transfer to a calibration window preceding the test window in time. Operators should recalibrate on a recent trusted window.

Composition affects detection, not the disparity conclusion. Restoring the ICMP class leaves the gap ordering unchanged but lowers TPR, concentrated in pooled-threshold variants (GlobalCP 0.9524→0.8891; the GlobalCal ablation falls to 0.7850). Under held-out attacks device-wise calibration is better on both axes: the device-level paired TPR difference over GlobalCP is +0.081, 95% device-cluster bootstrap interval $[+0.012, +0.139]$, and F1 +0.045, $[+0.019, +0.073]$.

Contaminated calibration windows. Injecting attacks into the benign calibration set at 0/1/3/5/10% affects both datasets in the same way: 1% is largely absorbed (Edge-IIoTset FPR 0.0284→0.0207 at unchanged TPR; N-BaIoT 0.0284→0.0201 at TPR 1.000), but from 3%—the target tail probability itself—the threshold is pulled into the attack score range and detection collapses (Edge-IIoTset TPR 0.9336 at 3%, 0.5753 at 5%; N-BaIoT 0.7449 and 0.5366). The apparent FPR improvement is threshold over-suppression, not a reliability gain. Duplicate removal on N-BaIoT leaves the core result intact (DeviceCP MaxGap 0.0138 ± 0.0055 vs. GlobalCP 0.2024 ± 0.0067).

Table 4. Adapter ablation at 50 rounds over three seeds, reference FedAvg+DeviceCP. "Cons." = the per-seed paired MaxGap difference against the reference has the same sign in all three seeds. N-BaIoT $F1_{arg}$ gives the three-seed mean and, in brackets, the worst seed.

	Edge-IIoTset			N-BaIoT		
Configuration	MaxGap↓	TPR_{conf}	Cons.	MaxGap↓	$F1_{arg}$ [worst seed]	Cons.
FedAvg+DeviceCP (no adapter)	0.0207	0.9565	—	0.0141	0.9995 [0.9994]	—
adapter rank 4	0.0204	0.9535	no	0.0137	0.9817 [**0.9464**]	no
adapter rank 8	0.0123	0.9562	yes	0.0128	0.9995 [0.9995]	no
adapter rank 16 (main configuration)	0.0174	0.9575	**no**	0.0094	0.9817 [**0.9464**]	yes
adapter rank 32	0.0123	0.9589	yes	0.0134	0.9995 [0.9994]	no
rank 16, early placement	**0.0090**	0.9563	yes	—	—	—
rank 16, adapter LR ×10	0.0159	0.9555	yes	0.0103	0.9817 [0.9463]	yes
rank 16, frozen first half	0.0136	0.9583	yes	0.0123	0.9817 [**0.9464**]	no

5.10 Adapter Ablation and Statistical Treatment

We ablate the adapter along all three axes at full scale (50 rounds, three seeds) with FedAvg+DeviceCP as reference; "early" placement inserts it after the first encoder block (256-d) rather than the encoder output (128-d).

Table 4 shows that the main rank 16 configuration is not a safe choice: it is the only rank whose N-BaIoT MaxGap improvement is consistent across seeds, yet it is also one of the two ranks that collapse (seed 44 argmax F1 = 0.9464), and on Edge-IIoTset it flips sign across seeds (−0.0003, +0.0037, −0.0130). The collapse is a rank-dependent optimization failure rather than seed noise: it reproduces at ranks 4 and 16, never at 8 or 32, and survives both a 10× adapter learning rate and freezing the adapter for the first 25 rounds. No configuration changes the headline conclusion—the best setting (early placement, MaxGap 0.0090) improves on the reference by 0.012 absolute, an order of magnitude below the 0.208 separating device-wise from global calibration, while doubling the local parameter count. Across the five non-held-out protocols in Table 3, the paired difference is consistently signed only on the primary random split (−0.005); chronological seed differences (+0.123, +0.040, −0.012) are sign-unstable and mostly favor no adapter.

Statistical treatment. Devices form a fixed fleet evaluated by all methods, so an unpaired test over device records would mismatch the design and inflate the effective sample size. We report per-seed paired differences, and for device-level metrics a percentile bootstrap resampling devices, which is descriptive for this fixed fleet rather than inference about a device population; fleet-level metrics have one value per seed, so we give all three values and their range without claiming an interval. Under this treatment the disparity claim holds and the adapter claim does not: the DeviceCP-minus-GlobalCP MaxGap difference is negative in every seed of every protocol, with reduction magnitudes from 0.039 to 0.298, while the device-level paired FPR difference between them is indistinguishable from zero (mean +0.003, interval spanning zero)—which is the point, since only the per-device distribution separates the two strategies. PFC-FedIDS minus FedAvg+DeviceCP has device-level intervals spanning zero for FPR, TPR and F1 on both datasets.

6 Discussion

6.1 When Does Global Calibration Fail?

Global calibration fails when devices have shifted benign score distributions, and the failure is not visible in the average FPR: FedAvg+GlobalCP tracks the requested average target while assigning very different effective thresholds to different device populations. It buys no detection advantage in exchange—TPR_{conf}=0.9528 on Edge-IIoTset versus 0.9575 for PFC-FedIDS and 0.9565 for FedAvg+DeviceCP—so it is strictly worse on device-wise disparity at equal sensitivity. The FedPer comparison isolates why: model personalization alone leaves MaxFPR-Gap at 0.2518 ± 0.0462, close to FedAvg+GlobalCP's 0.2280 ± 0.0075, and only adding device-wise calibration brings it to 0.0087 ± 0.0005. Model and threshold personalization are separate levers, and neither substitutes for the other. The effect scales with fleet heterogeneity: on N-BaIoT (all consumer devices) global calibration produces 8.8–13.8% gaps, while on Edge-IIoTset (heart-rate monitors through Modbus controllers) it produces 22.8–25.2%. If a deployment spans consumer, industrial and medical sensors, a single global alarm threshold is unsafe, and device-wise calibration should be the default—subject to the recalibration requirement established in §5.9.

6.2 Limitations and Future Work

Adapter effectiveness. Device-wise conformal calibration is the mechanism that delivers device-level false-alarm control; the adapter is optional and, on these backbones, not beneficial. Our ablation identifies the seed-44 collapse to argmax F1 0.9464 as rank-dependent (ranks 4 and 16, never 8 or 32); it survives both a 10× adapter learning rate and freezing the adapter for 25 rounds, so it is an optimization failure rather than seed noise. Whether adapters help in deeper backbones is untested here and is not a reason to keep them in the critical path.

Benchmark and preprocessing protocol. On Edge-IIoTset, clients use real sensor-specific benign traffic but synthetically assigned attack traces, so our evidence concerns benign false-alarm heterogeneity rather than real per-device attack exposure; the chronological protocol removes the time-ordering defect but not this one. Our partition covers 13 of the 14 released attack classes for the parsing reason documented in §5.1, and both datasets are balanced between benign and attack traffic, so F1 and precision do not transfer to operational base rates—FPR and TPR do not depend on class priors and carry our conclusions. Standardization is fitted on pooled training partitions; local or federated normalization remains future work.

Exchangeability and scope. Section 5.9 shows the per-device bound is not robust: under a chronological split combined with duplicate removal, one sensor reaches 13.4% FPR against a 3% target. Periodic recalibration on a recent trusted window, or adaptive conformal prediction [16], is therefore required for deployment rather than optional. Our results are also confined to 9–10 clients

on two tabular benchmarks with synchronous participation; larger fleets, unseen devices joining after calibration, and asynchronous participation are open.

Theoretical statement. Eq. (2) uses a one-sided device-wise FPR constraint matching the marginal per-device upper bound in Eq. (8), $\Pr(s(X) > q_i \mid Y = 0) \leq \alpha$, with q_i given by the finite-sample rule of Eq. (6) and alarms raised only when $s(x) > q_i$. The bound is marginal, not conditional, and does not imply simultaneous control across devices; with discrete scores or heavy ties the empirical FPR can fall well below α, which costs detection power. Small measured FPR gaps should therefore not be read as fleet-wide reliability.

7 Conclusion

We studied benign-only device-wise conformal calibration for heterogeneous federated IoT intrusion detection. On N-BaIoT and Edge-IIoTset, global calibration matches the average FPR while producing 22.8–25.2% device-level gaps; device-wise calibration reduces this gap by over 13× under the record-level random protocol while preserving detection (TPR_{conf}=0.9575 vs. 0.9528). The FedPer comparison shows that threshold personalization remains necessary with model personalization, and calibration needs no labeled attacks. The local adapter is unnecessary and unsafe at the reported rank. Under chronological, duplicate-free evaluation, however, the advantage shrinks to 1.6× and one sensor reaches 4.5× its target FPR. Deployment therefore requires periodic recalibration rather than a one-off guarantee.

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